THE THIRD LARGEST COSET LEADER MODULO q^m+1: A COUNTEREXAMPLE TO A CONJECTURE OF LIU, LI, GUO AND SONG

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Abstract. The largest coset leaders of q-cyclotomic cosets modulo n determine the largest designed distances of BCH codes of length n. For $n = q^m + 1$ with q a power of two and $m = 2^r t + 2^{r-1}$ ($r \geq 2$, $t \geq 1$), Liu, Li, Guo and Song proved a formula for the largest coset leader δ_1 and conjectured formulas for the second and third largest ones (Discrete Math. 2023, Conjecture 25). We show that the conjecture is false: for $q = 4$ and $m = 12$ the conjectured third largest coset leader 5875062 is not a coset leader at all, since $5875062 \cdot 4^6 \equiv 5724774 \pmod{4^{12}+1}$. This is verified in Lean 4. In computations with $q \in \{2, 4, 8, 16\}$, $2 \leq r \leq 4$ and $1 \leq t \leq 3$, the conjectured δ_2 was a coset leader in every case and the second largest leader in every case we could check, while the conjectured δ_3 failed exactly when $t = 1$ and $r \geq 3$, that is, for $m = 3 \cdot 2^{r-1} \geq 12$; in each of these cases it is not even a coset leader. For $q = 2$ and $m = 12$ the conjectured value also contradicts a theorem of Liu, Li, Fu, Lu and Rao. In every case we could check, the third largest coset leader for $t = 1$ equals $\delta_2 - (q-1)\, q^{2^{r-2}+1} \prod_{j=1}^{r-2} (q^{2^j} - 1)$, which we propose as a corrected conjecture.

1. Introduction

Let q be a prime power and n a positive integer coprime to q. The q-cyclotomic coset of x modulo n is $C_x = \{xq^i \bmod n : i \geq 0\}$, and its smallest element is its *coset leader*. The largest coset leaders determine the largest designed distances, and hence the Bose distances and dimensions, of narrow-sense BCH codes of length n over $\mathbb{F}_q$. They are known only for a few families of lengths. For lengths $n = q^m + 1$ (antiprimitive BCH codes) see [2, 1, 3] and the references there.

Liu, Li, Guo and Song [2] studied $n = q^m + 1$. In their Section 5.2, q is a power of 2. When m is not a power of two they write $m = 2^r t + 2^{r-1}$ with $r \geq 2$ and $t \geq 1$, and put

$$\Phi'(x,q) = \frac{q}{2}\left(q^{2^x}-1\right)\left(q^{2^{x-1}}-1\right)\cdots\left(q^{2^0}-1\right) \quad (x \geq 0), \qquad \Phi'(x,q) = \frac{q}{2} \quad (x < 0),$$

$$\delta_1 = \frac{n}{q^{2^{r-1}}+1}\,\Phi'(r-2,q), \qquad \delta_2 = \delta_1 - \frac{2\big(\delta_1 + \Phi'(r-1,q)\big)}{q^{2^r}},$$

$$\delta_3 = \begin{cases} \delta_2 - (q-1)q^2, & t = 1, \\ \delta_2 - 2\,\Phi'(r-1,q), & t > 1. \end{cases}$$

For $m = 2^r \geq 4$ they give other formulas for $\delta_1, \delta_2, \delta_3$, and for $m = 2$ explicit values. They proved that δ_1 is the largest coset leader [2, Theorem 24] and stated the following conjecture.

Conjecture 1.1 ([2, Conjecture 25])**.** *Let q be a power of 2, and let δ_2 and δ_3 be given as above. Then δ_2 and δ_3 are the second and third largest coset leaders modulo $n = q^m + 1$, respectively.*

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The authors are AI models: GPT-6-Astra (OpenAI) and Claude Opus 5.5 (Anthropic). Source files, scripts and Lean code: https://github.com/Kevin-OS7/math.

The paper [2] is about nonbinary codes, so $q > 2$ there. However, the remark following Conjecture 1.1 says that its statements also apply to $q = 2$, and that for $q = 2$ they were verified in [1] when $m \not\equiv 0 \pmod 8$.

2. The counterexample

Theorem 2.1. *Conjecture 1.1 is false. For $q = 4$ and $m = 12 = 2^3 \cdot 1 + 2^2$ (so $r = 3$ and $t = 1$), and $n = 4^{12} + 1 = 16777217$, the conjectured third largest coset leader $\delta_3 = 5875062$ is not a coset leader.*

Proof. Here $\Phi'(1, 4) = 2 \cdot 15 \cdot 3 = 90$ and $\Phi'(2, 4) = 2 \cdot 255 \cdot 15 \cdot 3 = 22950$. Hence

$$\delta_1 = \frac{16777217}{257} \cdot 90 = 65281 \cdot 90 = 5875290,$$
$$\delta_2 = 5875290 - \frac{2\,(5875290 + 22950)}{65536} = 5875290 - 180 = 5875110,$$

with the division exact, and $\delta_3 = \delta_2 - 3 \cdot 16 = 5875062$. But $5875062 \cdot 4^6 = 24064253952 = 1434 \cdot 16777217 + 5724774$. So $5724774 \in C_{5875062}$ and $5724774 < 5875062$, and 5875062 is not the smallest element of its coset. □

The parameters $q = 4 > 2$, $r = 3$, $t = 1$ lie in the nonbinary setting of [2]. A direct search shows that the three largest coset leaders modulo $4^{12} + 1$ are 5875290, 5875110 and 5872230. Thus the formulas for δ_1 and δ_2 are correct here, and the third largest leader is 5872230.

3. Where the conjecture fails

We tested the branch $m = 2^r t + 2^{r-1}$ for $q \in \{2, 4, 8, 16\}$, $r \in \{2, 3, 4\}$ and $t \in \{1, 2, 3\}$, which gives 36 parameter sets. For each set we decided whether the conjectured $\delta_1, \delta_2, \delta_3$ are coset leaders. This takes at most $2m$ modular multiplications per value, because the order of q modulo $q^m + 1$ is $2m$. When $n < 10^{12}$ (15 parameter sets) we also found the three largest coset leaders, by scanning downwards from δ_1, which is the largest leader by [2, Theorem 24]. When $n < 10^8$ we checked independently that δ_1 is the largest leader, by scanning down from $\lfloor n/2 \rfloor$. This suffices because $q^m \equiv -1 \pmod n$, so every coset contains both x and $n - x$. The results are as follows.

- δ_1 and δ_2 are coset leaders in all 36 sets, and δ_2 is the second largest leader in all 15 sets where the top three were computed.
- The conjectured δ_3 is a coset leader in all 28 sets with $r = 2$ or $t \geq 2$, and it is the third largest leader in all 11 of them where the top three were computed.
- For $t = 1$ and $r \in \{3, 4\}$, i.e. $m \in \{12, 24\}$, the conjectured δ_3 is *not a coset leader* for any $q \in \{2, 4, 8, 16\}$ (8 sets).

We also tested the branch $m = 2^r$ for $m \in \{2, 4, 8, 16\}$ and $q \in \{2, 4, 8, 16\}$. The cases $q = 2$, $m \leq 4$ were excluded because the formulas give non-positive values there. In all 14 remaining cases the three conjectured values are coset leaders, and they are the three largest leaders in all 12 cases with $n < 10^{12}$. So, as far as we could test, Conjecture 1.1 fails only for $t = 1$ and $r \geq 3$.

Remark 3.1. For $q = 2$ and $m = 12$ the conjectured value is $\delta_3 = 717 - 4 = 713$. But $713 \cdot 2^9 \equiv 423 \pmod{4097}$, and Theorem 5.2 of [1] (the case $t = 1$ of $m = 8t + 4$) gives $\delta_3 = \delta_2 - 24 = 693$, in agreement with our computation. Hence, for $t = 1$, the statement in [2] that the conjectures were verified in [1] for $q = 2$ and $m \not\equiv 0 \pmod 8$ is inaccurate. The same family $m = 3 \cdot 2^v$ is the one in which Zhang and Cao [3, Remark 1] found that the formula for δ_4 in the companion Conjecture 20 of [2] (odd q) must be corrected.

For $t = 1$ the data suggest the following correction. For $r = 2$ it reduces to the conjectured $\delta_2 - (q-1)q^2$, and for $q = 2$, $r = 3$ it gives $\delta_2 - 24$, as in [1, Theorem 5.2].

Conjecture 3.2. *Let q be a power of 2, $r \geq 2$ and $m = 2^r + 2^{r-1}$. Then the third largest coset leader modulo $q^m + 1$ is*

$$\delta_2 - (q-1)\, q^{2^{r-2}+1} \prod_{j=1}^{r-2} (q^{2^j} - 1).$$

We verified Conjecture 3.2 for $q \in \{2, 4, 8, 16\}$ with $r \in \{2, 3\}$, and for $(q, r) = (2, 4)$, $(4, 4)$ and $(2, 5)$. In each of these cases an exhaustive search shows that no coset leader lies strictly between δ_2 and δ_1, so that δ_2 is the second largest leader, and that the largest leader below δ_2 is the predicted value. For $(q, r) = (8, 4)$ the predicted value is a coset leader, but the range to be searched (about $5.9 \cdot 10^{10}$ values) was too large.

All computations use exact integer arithmetic and are made by a self-contained Python script, `conj25_check.py`, which accompanies the source files together with its output. The whole run takes about half a minute on one core.

4. Formal verification

The conjecture and the counterexample were formalized in Lean 4 (version 4.33.1) with Mathlib. The coset leader property is stated as

```
def IsCosetLeader (q n x : ℕ) : Prop :=
  x < n ∧ ∀ i : ℕ, x ≤ (x * q ^ i) % n
```

For the moduli considered here this is equivalent to minimality in the coset, since $\gcd(q, n) = 1$. The functions Φ', δ_1, δ_2 and δ_3 are transcribed from [2, Sect. 5.2], and "x is the k-th largest leader" is formalized as "x is a leader and exactly $k-1$ leaders are larger than x". We prove two theorems. The first is the negation of the branch $m = 2^r t + 2^{r-1}$ of Conjecture 1.1, quantified over all powers of two q and all $r \geq 2$, $t \geq 1$; the full conjecture implies this branch. The second is the explicit statement of Theorem 2.1. Both depend only on the axioms `propext`, `Classical.choice` and `Quot.sound`. The computations of Section 3 are not formalized.

Author contributions. A Claude Opus 5.5 agent surveying open problems in coding theory found the counterexample. GPT-6-Astra checked it against the exact statement and conventions of [2], checked the status of the conjecture, formalized it and wrote the Lean proof of its negation. Claude Opus 5.5 checked the transcription of the formulas, carried out the computations of Section 3, found the discrepancy with [1], proposed Conjecture 3.2, added the explicit Lean theorem and wrote the manuscript. A human operator initiated the project and provided computing resources; the mathematics, the formal proofs and the text were produced by the models.

References

[1] Y. Liu, R. Li, Q. Fu, L. Lu and Y. Rao, *Some binary BCH codes with length $n = 2^m + 1$*, Finite Fields Appl. **55** (2019), 109–133; arXiv:1803.02731.

[2] Y. Liu, R. Li, L. Guo and H. Song, *Dimensions of nonbinary antiprimitive BCH codes and some conjectures*, Discrete Math. **346** (2023), no. 9, 113496; arXiv:1712.06842.

[3] H. Zhang and X. Cao, *Dimensions of some LCD BCH codes*, arXiv:2305.06508 (2023).